

Develop a GUI-based banking application that allows users to perform operations such as fund transfer, balance inquiry, and viewing transaction history. Use Swing components and event listeners to manage user interactions securely. Implement input validation for sensitive data like account number and amount. Ensure proper error handling for invalid transactions. Apply suitable layout managers to create a responsive and user-friendly interface. Analyze system responsiveness during multiple operations and suggest improvements. Demonstrate how secure input handling enhances reliability.

Full Source Code

```
import javax.swing.*;
import javax.swing.table.*;
import java.awt.*;
import java.awt.event.*;
import java.text.SimpleDateFormat;
import java.util.*;

public class OnlineBankingInterface extends JFrame {

    static class Account {
        String id, name, pin;
        double balance;
        ArrayList<String[]> history = new ArrayList<>();
        Account(String id, String name, String pin, double balance) {
            this.id=id; this.name=name; this.pin=pin; this.balance=balance;
            addHistory("Credit", balance, "Account Opened", balance);
        }
        void addHistory(String type, double amount, String desc, double bal) {
            String date = new SimpleDateFormat("dd-MM-yyyy HH:mm").format(new
Date());
            history.add(new String[]{date,type,
                String.format("%.2f",amount),desc,String.format("%.2f",bal)}});
        }
    }

    private final HashMap<String,Account> accounts = new HashMap<>();
    private Account currentAccount = null;
    private final CardLayout cardLayout = new CardLayout();
    private final JPanel cardPanel = new JPanel(cardLayout);
    private final JTextField tfAccLogin = new JTextField(16);
    private final JPasswordField pfPinLogin = new JPasswordField(16);
    private final JLabel lblLoginError = new JLabel(" ");
    private final JLabel lblWelcome = new JLabel();
    private final JLabel lblBalance = new JLabel();
    private final JLabel lblAccId = new JLabel();
    private final JTextField tfToAcc = new JTextField(16);
    private final JTextField tfTransAmt = new JTextField(16);
    private final JTextField tfTransDesc = new JTextField(16);
    private final JLabel lblTransMsg = new JLabel(" ");
    private final DefaultTableModel histModel;
    private final JTable histTable;
    private final JLabel statusBar = new JLabel(" Please log in.");

    public OnlineBankingInterface() {
        super("SecureBank - Online Banking");
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setSize(920, 580);
        setLocationRelativeTo(null);
        accounts.put("1001",new Account("1001","Azeez Najeeb","1234",45000));
        accounts.put("1002",new Account("1002","Priya Sharma", "2345",12500));
        accounts.put("1003",new Account("1003","Ravi Kumar", "3456",88000));
        String[] cols={"Date & Time","Type","Amount (Rs)","Description","Balance
(Rs)"};
        histModel = new DefaultTableModel(cols, 0) {
            public boolean isCellEditable(int r,int c){return false;}
        };
        histTable = new JTable(histModel);
```

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        styleHistTable();
        cardPanel.add(buildLoginScreen(), "LOGIN");
        cardPanel.add(buildDashboard(), "DASH");
        setLayout(new BorderLayout());
        add(buildTopBar(), BorderLayout.NORTH);
        add(cardPanel, BorderLayout.CENTER);
        add(buildStatus(), BorderLayout.SOUTH);
        cardLayout.show(cardPanel, "LOGIN");
        setVisible(true);
    }

    private JPanel buildTopBar() {
        JPanel p = new JPanel(new BorderLayout());
        p.setBackground(new Color(20, 60, 100));
        p.setPreferredSize(new Dimension(0, 50));
        JLabel lbl = new JLabel(" SecureBank - Online Banking Portal");
        lbl.setFont(new Font("Segoe UI", Font.BOLD, 18));
        lbl.setForeground(Color.WHITE);
        p.add(lbl, BorderLayout.WEST);
        JButton btnLogout = new JButton("Logout");
        btnLogout.setBackground(new Color(192, 57, 43));
        btnLogout.setForeground(Color.WHITE);
        btnLogout.setFocusPainted(false);
        btnLogout.addActionListener(e -> logout());
        JPanel r = new JPanel(new FlowLayout(FlowLayout.RIGHT));
        r.setOpaque(false); r.add(btnLogout);
        p.add(r, BorderLayout.EAST); return p;
    }

    private JPanel buildLoginScreen() {
        JPanel outer = new JPanel(new GridBagLayout());
        outer.setBackground(new Color(236, 240, 241));
        JPanel card = new JPanel(new GridLayout(0, 1, 8, 8));
        card.setBorder(BorderFactory.createCompoundBorder(
            BorderFactory.createLineBorder(new Color(174, 182, 191), 1),
            BorderFactory.createEmptyBorder(30, 40, 30, 40));
        card.setBackground(Color.WHITE);
        card.setPreferredSize(new Dimension(380, 340));
        JLabel title = new JLabel("Account Login", SwingConstants.CENTER);
        title.setFont(new Font("Segoe UI", Font.BOLD, 20));
        title.setForeground(new Color(20, 60, 100));
        lblLoginError.setForeground(Color.RED);
        JButton btnLogin = new JButton("Login");
        btnLogin.setBackground(new Color(20, 60, 100));
        btnLogin.setForeground(Color.WHITE);
        btnLogin.addActionListener(e -> doLogin());
        card.add(title);
        card.add(new JLabel("Account Number")); card.add(tfAccLogin);
        card.add(new JLabel("PIN")); card.add(pfPinLogin);
        card.add(lblLoginError);
        card.add(btnLogin);
        JLabel hint = new JLabel("Demo: 1001/1234 1002/2345 1003/3456",
            SwingConstants.CENTER);
        hint.setFont(new Font("Segoe UI", Font.ITALIC, 11));
        hint.setForeground(Color.GRAY);
        card.add(hint);
        outer.add(card); return outer;
    }

    private JPanel buildDashboard() {
        JPanel p = new JPanel(new BorderLayout(8, 8));
        p.setBorder(BorderFactory.createEmptyBorder(10, 10, 10, 10));
        p.setBackground(new Color(245, 247, 250));
        JPanel infoCard = new JPanel(new FlowLayout(FlowLayout.LEFT, 30, 10));
        infoCard.setBackground(new Color(20, 60, 100));
        lblWelcome.setFont(new Font("Segoe UI", Font.BOLD, 16));
        lblWelcome.setForeground(Color.WHITE);
        lblAccId.setFont(new Font("Segoe UI", Font.PLAIN, 13));
        lblAccId.setForeground(new Color(174, 214, 241));
        lblBalance.setFont(new Font("Segoe UI", Font.BOLD, 18));
        lblBalance.setForeground(new Color(88, 214, 141));
        infoCard.add(lblWelcome); infoCard.add(lblAccId); infoCard.add(lblBalance);
        p.add(infoCard, BorderLayout.NORTH);
        JTabbedPane tabs = new JTabbedPane();
        tabs.setFont(new Font("Segoe UI", Font.BOLD, 13));
    }

```

```

tabs.addTab("Fund Transfer", buildTransferPanel());
tabs.addTab("Balance Inquiry", buildBalancePanel());
tabs.addTab("Transaction History", buildHistoryPanel());
p.add(tabs, BorderLayout.CENTER); return p;
}

```

Expected Output / GUI Description

 **SecureBank — Online Banking Portal**
Logout

Account Holder

Azeez Najeed

Account No.

1001

Available Balance

₹ 44,500.00

 Fund Transfer

 Balance Inquiry

 Transaction History

 **Transfer Funds**

Beneficiary Account No. *

1002

Amount (₹) *

500

Description

Monthly payment

 ₹500.00 transferred to Priya Sharma successfully.

Transfer Now

 Transfer of ₹500.00 to account 1002 completed successfully.